

Assignment 2 – The LP Model

1. Back Savers

Back Savers produces two types of backpacks: the Collegiate and the Mini. The goal is to determine the number of each backpack that should be produced each week in order to maximize weekly profit while staying within available fabric, labor, and sales limitations.

a. Decision Variables

Let:

C = number of Collegiate backpacks produced per week

M = number of Mini backpacks produced per week

b. Objective Function

Each Collegiate backpack generates a profit of \$32, and each Mini backpack generates a profit of \$24.

Therefore, the objective is:

$$\text{Maximize } Z = 32C + 24M$$

where Z represents total weekly profit.

c. Constraints

Nylon Fabric Constraint

Back Savers receives 5,000 square feet of nylon each week.

Each Collegiate requires 3 square feet, and each Mini requires 2 square feet.

$$3C + 2M \leq 5000$$

Collegiate Sales Constraint

The company can sell at most 1,000 Collegiate backpacks per week.

$$C \leq 1000$$

Mini Sales Constraint

The company can sell at most 1,200 Mini backpacks per week.

$$M \leq 1200$$

Labor Constraint

Back Savers has 35 employees who each work 40 hours per week.

Total labor available is:

$$35(40) = 1400 \text{ hours}$$

Because the production times are given in minutes, this can be converted to:

$$1400(60) = 84,000 \text{ minutes}$$

Each Collegiate requires 45 minutes of labor, while each Mini requires 40 minutes.

Therefore:

$$45C + 40M \leq 84,000$$

Nonnegativity Constraints

$$C, M \geq 0$$

d. Full Mathematical Formulation

$$\text{Maximize } Z = 32C + 24M$$

Subject to:

$$3C + 2M \leq 5000$$

$$C \leq 1000$$

$$M \leq 1200$$

$$45C + 40M \leq 84,000$$

$$C, M \geq 0$$

2. Weigelt Corporation

The Weigelt Corporation has three plants that can manufacture three sizes of a new product: large, medium, and small. Management wants to determine how much of each size should be produced at each plant in order to maximize daily profit while satisfying production capacity, storage, sales, and workforce-related requirements.

a. Decision Variables

Let:

L_i = number of large units produced per day at plant i

M_i = number of medium units produced per day at plant i

S_i = number of small units produced per day at plant i

where:

$$i = 1, 2, 3$$

Therefore, the nine production decision variables are:

$$L_1, M_1, S_1, L_2, M_2, S_2, L_3, M_3, S_3$$

Also define:

p = common percentage of excess production capacity used at each plant

where:

$$0 \leq p \leq 1$$

b. Linear Programming Model

Objective Function

The profit per unit is:

- Large: \$420
- Medium: \$360
- Small: \$300

Therefore, total daily profit is:

$$\text{Maximize } Z = 420(L_1 + L_2 + L_3) + 360(M_1 + M_2 + M_3) + 300(S_1 + S_2 + S_3)$$

Production Capacity and Equal Utilization Constraints

Plant 1 can produce at most 750 units per day. Plant 2 can produce 900 units per day, and Plant 3 can produce 450 units per day.

Management requires each plant to use the same percentage of its available excess capacity.

Using p to represent this common percentage:

$$L_1 + M_1 + S_1 = 750p$$

$$L_2 + M_2 + S_2 = 900p$$

$$L_3 + M_3 + S_3 = 450p$$

and:

$$0 \leq p \leq 1$$

These constraints guarantee that all three plants use the same percentage of their available excess production capacity.

Storage Constraints

Each large unit requires 20 square feet of storage, each medium unit requires 15 square feet, and each small unit requires 12 square feet.

Plant 1 has 13,000 square feet available:

$$20L_1 + 15M_1 + 12S_1 \leq 13,000$$

Plant 2 has 12,000 square feet available:

$$20L_2 + 15M_2 + 12S_2 \leq 12,000$$

Plant 3 has 5,000 square feet available:

$$20L_3 + 15M_3 + 12S_3 \leq 5,000$$

Sales Constraints

Demand for large units is limited to 900 units per day:

$$L_1 + L_2 + L_3 \leq 900$$

Demand for medium units is limited to 1,200 units per day:

$$M_1 + M_2 + M_3 \leq 1200$$

Demand for small units is limited to 750 units per day:

$$S_1 + S_2 + S_3 \leq 750$$

Nonnegativity Constraints

$$L_i, M_i, S_i \geq 0 \quad i = 1, 2, 3$$

$$0 \leq p \leq 1$$

Full Mathematical Formulation

$$\text{Maximize } Z = 420(L_1 + L_2 + L_3) + 360(M_1 + M_2 + M_3) + 300(S_1 + S_2 + S_3)$$

Subject to:

$$L_1 + M_1 + S_1 = 750p$$

$$L_2 + M_2 + S_2 = 900p$$

$$L_3 + M_3 + S_3 = 450p$$

$$20L_1 + 15M_1 + 12S_1 \leq 13,000$$

$$20L_2 + 15M_2 + 12S_2 \leq 12,000$$

$$20L_3 + 15M_3 + 12S_3 \leq 5,000$$

$$L_1 + L_2 + L_3 \leq 900$$

$$M_1 + M_2 + M_3 \leq 1200$$

$$S_1 + S_2 + S_3 \leq 750$$

$$L_i, M_i, S_i \geq 0 \quad i = 1, 2, 3$$

$$0 \leq p \leq 1$$